

ERASTEEL TECHNICAL DATASHEET: ASP 2190

Application Family: Specialties | Standards: Erasteel Ultra-High Cobalt Specialty PM

1. Grade Overview & Characteristics

ASP 2190 is a revolutionary specialty grade containing 29% Cobalt, specifically engineered for extreme hot hardness, thermal fatigue resistance, and PVD coating adhesion in severe dry gear cutting operations.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	0.74 - 0.82	4.0 - 4.4	2.7 - 3.1	2.7 - 3.1	1.0 - 1.2	1.126.5 to 2.05	(Nb) for ultra-high temperature grain

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	68.0 - 70.0 HRC	Machinability	3.5/10
Max Reachable Hardness	70.0 HRC	Wear Resistance	9.0/10
Annealed Delivery State	Max 280-350 HB	Toughness	4.5/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	10.0/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	4.0/10

4. Industrial Applications

Dry high-speed gear skiving, heavy-duty dry gear hobbing, aerospace turbine blade machining, extreme temperature wear parts.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1190-1215°C. Triple/quadruple tempering at 560-580°C. Working hardness up to 70 HRC.